Nathaniel Cavazos

Mechanical Engineering Student

Mechanical Engineering student with experience in CAD design, structural analysis, robotics, and MATLAB-based simulation. Interested in aerospace systems, controls engineering, and propulsion.

CONTACT

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U.S. Citizen | Eligible to Obtain a U.S. Security Clearance

EDUCATION

California State University, Long Beach | Long Beach, CA Expected May 2028 | GPA: 3.57

B.S. Mechanical Engineering

Relevant Coursework: Engineering Design, Statics, Dynamics, Deformable Bodies, Fluid Mechanics, Numerical Methods, & Computer Methods.

ACTIVITIES

Guided Navigation Control Engineer | Beach Launch Team Aug 2026 - Present

- Collaborating with students to develop a new dual-deployment parachute recovery system for the Theseus rocket.
- Developing recovery system components in Onshape in preparation for future testing.

Clubs: SASE, Pi Tau Sig

ENGINEERING PROJECTS

6-DOF Robotic Arm | V1: Design & Simulation | Personal Sep 2026 - Oct 2026

- Designed a 6-DOF robotic arm in SOLIDWORKS, evaluating component tolerances and refining joint assemblies to improve mechanical fit.
- Performed structural analysis and topology optimization in ANSYS to improve the strength-to-weight ratio while maintaining critical mounting features.
- Developed a 6-DOF kinematic model in MATLAB, performing Jacobian analysis and simulating coordinated motion of joints J1–J4 using Robotics System Toolbox.

Engineering Instrumentation & Calibration | Academic Oct 2026

- Analyzed Bourdon gauge and load-cell calibration data to evaluate measurement accuracy, sensitivity, and hysteresis.
- Applied linear regression and correlation analysis to ealuate sensor response and identify measuirement deviations

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WORK EXPERIENCE

Cafeteria Team Member | Delano High School, Delano, CA Apr 2023 - Jan 2025

- Collaborated with a team of 15 staff to serve 1,200+ high school students daily in a high-volume, fast-paced environment.
- Coordinated food preparation and distribution to ensure timely service during peak lunch periods.
- Led a team of 4 to manage inventory restocking and food preparation, maintaining continuous and organized service flow.

TECHNICAL SKILLS

CAD/CAE: Solidworks, Siemens NX, ANSYS

Controls & Simulation: MATLAB, Simulink, Robotics System Toolbox

Embedded: SCADA, UART, I2C, Boolean Logic

Engineering Software: KiCad, Ignition, Common Components Workbench, Excel